



SYSTEM: 7ep

system: "7ep"

services:

- name: "Authentication Service"

responsibilities:

- "Register librarians"
- "Validate librarian credentials"
- "Enforce password strength rules"
- "Manage and validate user sessions"
- "Report authentication events for auditing"

dependencies:

- "Audit Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Book Catalog Service"

responsibilities:

- "Register books in collection"
- "Prevent duplicate book registration"
- "Delete registered books"
- "Consult books by identifier or title"
- "List all books"
- "List books available for loan"
- "Coordinate loan link removal upon book deletion"
- "Report catalog operations for auditing"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/book-catalog-service"

- name: "Reader Management Service"

responsibilities:

- "Register readers eligible for loans"
- "Prevent duplicate reader registration"
- "Delete registered readers"
- "Consult readers by identifier or name"
- "List all registered readers"
- "Coordinate loan link removal upon reader deletion"
- "Report reader operations for auditing"

dependencies:

- "Loan Management Service"

- "Audit Service"

suggested_endpoints:

- "/api/reader-management-service"

- name: "Loan Management Service"

responsibilities:

- "Register book loans"

- "Record loan dates"

- "Ensure books are loaned to only one reader at a time"

- "Validate reader eligibility"

- "Validate book availability and eligibility"

- "Remove loan links upon book or reader deletion"

- "Report loan operations for auditing"

dependencies:

- "Reader Management Service"

- "Book Catalog Service"

- "Audit Service"

suggested_endpoints:

- "/api/loan-management-service"

- name: "Environment Management Service"

responsibilities:

- "Clear system data and base structure"

- "Recreate base structure"

- "Restore clean environment state"

- "Coordinate reset operations across functional areas"

dependencies:

- "Authentication Service"

- "Book Catalog Service"

- "Reader Management Service"

- "Loan Management Service"

- "Audit Service"

suggested_endpoints:

- "/api/environment-management-service"

- name: "Demo Utility Service"

responsibilities:

- "Sum two integers"

- "Calculate Fibonacci sequence values"

- "Calculate Ackermann function results"

- "Validate mathematical input format"

- "Provide catalog for consulting and testing endpoints"

dependencies:

- ""

suggested_endpoints:

- "/api/demo-utility-service"

- name: "Audit Service"

responsibilities:

- "Record relevant operations from registration authentication consultation and loan flows"
- "Maintain decoupled central event records"
- "Support traceability failure tracking and problem analysis"

dependencies:

- ""

suggested_endpoints:

- "/api/audit-service"

interactions:

- from: "Authentication Service"
to: "Audit Service"
- from: "Book Catalog Service"
to: "Loan Management Service"
- from: "Book Catalog Service"
to: "Audit Service"
- from: "Reader Management Service"
to: "Loan Management Service"
- from: "Reader Management Service"
to: "Audit Service"
- from: "Loan Management Service"
to: "Reader Management Service"
- from: "Loan Management Service"
to: "Book Catalog Service"
- from: "Loan Management Service"
to: "Audit Service"
- from: "Environment Management Service"
to: "Authentication Service"
- from: "Environment Management Service"
to: "Book Catalog Service"
- from: "Environment Management Service"
to: "Reader Management Service"
- from: "Environment Management Service"
to: "Loan Management Service"
- from: "Environment Management Service"
to: "Audit Service"

SYSTEM: acmeair

system: "acmeair"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate customer credentials"
- "Create usage sessions"
- "Validate active sessions"
- "Invalidate sessions on logout"
- "Manage session expiration"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Flight Search Service"

responsibilities:

- "Manage registered airports and routes"
- "Search one-way and round-trip flights"
- "Provide flight schedules and trip segments"
- "Provide reference fare values"

dependencies:

- ""

suggested_endpoints:

- "/api/flight-search-service"

- name: "Booking Service"

responsibilities:

- "Create one-way and round-trip bookings"
- "Generate unique booking identifiers"
- "Record booking dates"
- "List completed customer bookings"
- "Cancel selected bookings"

dependencies:

- "Authentication Service"
- "Flight Search Service"

suggested_endpoints:

- "/api/booking-service"

- name: "Customer Profile Service"

responsibilities:

- "Consult customer profile"
- "Update customer contact and address data"
- "Validate current password for profile changes"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-profile-service"

- name: "Loyalty Service"

responsibilities:

- "Consult customer loyalty program status"

- "Present annual accumulated miles"

- "Present total accumulated miles"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/loyalty-service"

- name: "Environment Administration Service"

responsibilities:

- "Load and reload application operational data"

- "Generate sample customers and operational test records"

- "Recreate flights"

dependencies:

- "customers"

suggested_endpoints:

- "/api/environment-administration-service"

interactions:

- from: "Booking Service"

- to: "Authentication Service"

- from: "Booking Service"

- to: "Flight Search Service"

- from: "Customer Profile Service"

- to: "Authentication Service"

- from: "Loyalty Service"

- to: "Authentication Service"

- from: "Environment Administration Service"

- to: "customers"

SYSTEM: Cargo-Tracker

system: "Cargo-Tracker"

services:

- name: "Cargo Management Service"

responsibilities:

- "Register new cargo with origin"

dependencies:

- "destination"

suggested_endpoints:

- "/api/cargo-management-service"

- name: "Route Planning Service"

responsibilities:

- "Retrieve and present candidate routes matching origin"

dependencies:

- "destination"

suggested_endpoints:

- "/api/route-planning-service"
- name: "Logistics Event Service"
- responsibilities:
 - "Register logistics events of type reception"
- dependencies:
 - "load"
- suggested_endpoints:
 - "/api/logistics-event-service"
- name: "Cargo Tracking Service"
- responsibilities:
 - "Provide cargo consultation by tracking identifier"
 - "Consolidate current cargo situation"
- dependencies:
 - "last known location"
- suggested_endpoints:
 - "/api/cargo-tracking-service"
- name: "Integration Service"
- responsibilities:
 - "Receive logistics events via operational interface"
- dependencies:
 - "integration service"
- suggested_endpoints:
 - "/api/integration-service"

interactions:

- from: "Cargo Management Service"
- to: "destination"
- from: "Route Planning Service"
- to: "destination"
- from: "Logistics Event Service"
- to: "load"
- from: "Cargo Tracking Service"
- to: "last known location"
- from: "Integration Service"
- to: "integration service"

SYSTEM: DayTrader7

system: "DayTrader7"

services:

- name: "Authentication Service"
- responsibilities:
 - "Authenticate investor credentials"
 - "Manage active investor sessions"
 - "Process investor logout"

- "Enforce access control to restricted resources"

dependencies:

- "Investor Account Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Investor Account Service"

responsibilities:

- "Register new investors with registration validation"
- "Update investor profile details"
- "Maintain investor account financial data and cash balance"
- "Record access dates and login and logout counters"
- "Process balance debits and credits from trading operations"

dependencies:

- "Authentication Service"
- "Trading Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/investor-account-service"

- name: "Market Data Service"

responsibilities:

- "Provide individual and simultaneous asset quote queries"
- "Handle invalid asset queries gracefully"
- "Generate consolidated market summaries and performance indicators"
- "Update traded volume and market indicators from operations"
- "Apply configurable refresh intervals for market summary data"

dependencies:

- "Trading Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/market-data-service"

- name: "Trading Service"

responsibilities:

- "Create and manage lifecycle of buy and sell orders"
- "Apply fixed operation fees to trading orders"
- "Execute orders immediately or deferred according to simulation configuration"
- "Track order statuses and prevent duplicate completion"
- "Maintain investor order history and recent order summaries"
- "Emit alerts for completed orders"

dependencies:

- "Investor Account Service"
- "Portfolio Service"
- "Market Data Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/trading-service"

- name: "Portfolio Service"

responsibilities:

- "Maintain investor asset positions"
- "Register positions upon completed buy orders"
- "Remove positions upon completed sell orders"
- "Consult portfolio positions with market value and gain or loss"
- "Calculate consolidated financial position summary and accumulated results"
- "Initiate sell operations directly from portfolio positions"

dependencies:

- "Trading Service"
- "Market Data Service"
- "Investor Account Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/portfolio-service"

- name: "Simulation Administration Service"

responsibilities:

- "Configure simulation operational parameters and order processing modes"
- "Configure completed order alert settings"
- "Prepare and rebuild simulation database"
- "Reset simulation operational data"
- "Provide execution statistics and operational monitoring indicators"

dependencies:

- "Investor Account Service"
- "Trading Service"
- "Portfolio Service"
- "Market Data Service"

suggested_endpoints:

- "/api/simulation-administration-service"

interactions:

- from: "Authentication Service"
to: "Investor Account Service"
- from: "Investor Account Service"
to: "Authentication Service"
- from: "Investor Account Service"
to: "Trading Service"
- from: "Investor Account Service"
to: "Simulation Administration Service"
- from: "Market Data Service"
to: "Trading Service"
- from: "Market Data Service"

to: "Simulation Administration Service"

- from: "Trading Service"

to: "Investor Account Service"

- from: "Trading Service"

to: "Portfolio Service"

- from: "Trading Service"

to: "Market Data Service"

- from: "Trading Service"

to: "Simulation Administration Service"

- from: "Portfolio Service"

to: "Trading Service"

- from: "Portfolio Service"

to: "Market Data Service"

- from: "Portfolio Service"

to: "Investor Account Service"

- from: "Portfolio Service"

to: "Simulation Administration Service"

- from: "Simulation Administration Service"

to: "Investor Account Service"

- from: "Simulation Administration Service"

to: "Trading Service"

- from: "Simulation Administration Service"

to: "Portfolio Service"

- from: "Simulation Administration Service"

to: "Market Data Service"

SYSTEM: Jokul

system: "Jokul"

services:

- name: "Authentication Service"

responsibilities:

- "Register users with unique usernames"
- "Authenticate user login credentials"
- "Manage user sessions and logout"
- "Control and differentiate user profiles"
- "Validate user access and permissions"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Catalog Service"

responsibilities:

- "List movies with paginated navigation"

- "Present movie details and cast"
- "Manage movie categories and categorize movies"
- "Filter movies by category"
- "Register new movies with unique titles"
- "Delete movies individually or in batch"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Media Service"

responsibilities:

- "Upload movie video files"
- "Store and load video files from local repository"
- "Prepare and deliver video files for playback"
- "Provide movie download links"
- "Verify video file availability for registered movies"

dependencies:

- "Catalog Service"
- "Authentication Service"

suggested_endpoints:

- "/api/media-service"

interactions:

- from: "Catalog Service"
to: "Authentication Service"
- from: "Media Service"
to: "Catalog Service"
- from: "Media Service"
to: "Authentication Service"

SYSTEM: JpetStore

system: "JpetStore"

services:

- name: "Catalog Service"

responsibilities:

- "Browse virtual store catalog"
- "Organize products by animal categories"
- "Display product lists by category"
- "Display available items and item details"
- "Search products by keyword"
- "Validate search keyword input"

dependencies:

- "Inventory Service"

suggested_endpoints:

- "/api/catalog-service"
- name: "Customer Account Service"
- responsibilities:
 - "Register new customer profiles and credentials"
 - "Manage profile data and addresses"
 - "Configure customer preferences and favorite categories"
 - "Enable personalized product lists and promotional banners"
- dependencies:
 - "Authentication Service"
- suggested_endpoints:
 - "/api/customer-account-service"
- name: "Authentication Service"
- responsibilities:
 - "Authenticate user credentials"
 - "Manage customer sessions and logout"
 - "Enforce access control for protected operations"
- dependencies:
 - "Customer Account Service"
- suggested_endpoints:
 - "/api/authentication-service"
- name: "Shopping Cart Service"
- responsibilities:
 - "Manage shopping cart items"
 - "Consolidate duplicate items and update quantities"
 - "Calculate cart subtotal"
 - "Remove items from cart"
 - "Clear shopping cart"
- dependencies:
 - "Catalog Service"
 - "Inventory Service"
- suggested_endpoints:
 - "/api/shopping-cart-service"
- name: "Inventory Service"
- responsibilities:
 - "Track item stock availability"
 - "Indicate backorder status for unavailable items"
 - "Provide updated stock status"
 - "Register stock decrease upon order finalization"
- dependencies:
 - "Catalog Service"
- suggested_endpoints:
 - "/api/inventory-service"
- name: "Order Service"

responsibilities:

- "Coordinate checkout process"
- "Capture payment details and card selection"
- "Manage billing and delivery addresses"
- "Finalize orders and generate unique purchase identifiers"
- "Record order history and purchase details"
- "Retrieve customer order list and order details"
- "Enforce customer access restrictions on orders"

dependencies:

- "Authentication Service"
- "Customer Account Service"
- "Shopping Cart Service"
- "Inventory Service"

suggested_endpoints:

- "/api/order-service"

interactions:

- from: "Catalog Service"
to: "Inventory Service"
- from: "Customer Account Service"
to: "Authentication Service"
- from: "Authentication Service"
to: "Customer Account Service"
- from: "Shopping Cart Service"
to: "Catalog Service"
- from: "Shopping Cart Service"
to: "Inventory Service"
- from: "Inventory Service"
to: "Catalog Service"
- from: "Order Service"
to: "Authentication Service"
- from: "Order Service"
to: "Customer Account Service"
- from: "Order Service"
to: "Shopping Cart Service"
- from: "Order Service"
to: "Inventory Service"

SYSTEM: Pet-Clinic

system: "Pet-Clinic"

services:

- name: "Owner Service"

responsibilities:

- "Register owners with contact details"

- "Update registered owner information"
- "Search owners by last name prefix or broad search with pagination"
- "Retrieve detailed owner information"

dependencies:

- ""

suggested_endpoints:

- "/api/owner-service"

- name: "Animal Service"

responsibilities:

- "Register animals linked to an owner"
- "Update registered animal information"
- "Manage predefined animal types"
- "Validate animal birth dates and enforce unique animal names per owner"

dependencies:

- "Owner Service"

suggested_endpoints:

- "/api/animal-service"

- name: "Visit Service"

responsibilities:

- "Register visits with date and service description"
- "Suggest current date for new visits"
- "Associate visits with animal history and owner"
- "Maintain visit records and service history"
- "Display visit history of animals"

dependencies:

- "Animal Service"
- "Veterinarian Service"

suggested_endpoints:

- "/api/visit-service"

- name: "Veterinarian Service"

responsibilities:

- "List registered veterinarians and specialties in paginated and structured formats"
- "Automatically assign visits prioritizing veterinarians with lowest workload"
- "Calculate veterinarian workload and determine next expected visits"
- "Periodically disregard past visit assignments in workload calculation"
- "Optimize frequent veterinarian queries"

dependencies:

- "Visit Service"

suggested_endpoints:

- "/api/veterinarian-service"

interactions:

- from: "Animal Service"
- to: "Owner Service"

- from: "Visit Service"
to: "Animal Service"
- from: "Visit Service"
to: "Veterinarian Service"
- from: "Veterinarian Service"
to: "Visit Service"

SYSTEM: TNTConcept

system: "TNTConcept"

services:

- name: "Administration Service"

responsibilities:

- "Maintain support records including departments"

dependencies:

- "contract types"

suggested_endpoints:

- "/api/administration-service"

- name: "Identity and Access Service"

responsibilities:

- "Register and maintain users with personal"

dependencies:

- "contractual"

suggested_endpoints:

- "/api/identity-and-access-service"

- name: "Workforce Management Service"

responsibilities:

- "Maintain work agreements and vacation requests"

- "Control vacation request states and calculate balances"

dependencies:

- "Administration Service"

- "Identity and Access Service"

- "Notification Service"

suggested_endpoints:

- "/api/workforce-management-service"

- name: "Project and Work Tracking Service"

responsibilities:

- "Maintain projects linked to clients"

- "Register internal and external activities"

- "Classify activities and calculate worked hours"

- "Support evidence policies"

dependencies:

- "Identity and Access Service"

- "Workforce Management Service"

- "File Storage Service"
- "Notification Service"

suggested_endpoints:

- "/api/project-and-work-tracking-service"

- name: "CRM and Proposals Service"

responsibilities:

- "Maintain organizations"

dependencies:

- "contacts"

suggested_endpoints:

- "/api/crm-and-proposals-service"

- name: "Financial Management Service"

responsibilities:

- "Maintain accounts"

dependencies:

- "entries"

suggested_endpoints:

- "/api/financial-management-service"

- name: "Billing Service"

responsibilities:

- "Detail invoices and calculate totals"

dependencies:

- "taxes"

suggested_endpoints:

- "/api/billing-service"

- name: "Content and Document Management Service"

responsibilities:

- "Organize documents by categories and versions"

dependencies:

- "Identity and Access Service"
- "File Storage Service"

suggested_endpoints:

- "/api/content-and-document-management-service"

- name: "Commissioning Service"

responsibilities:

- "Register and manage orders and commissions"

dependencies:

- "Project and Work Tracking Service"
- "Workforce Management Service"
- "File Storage Service"
- "Notification Service"

suggested_endpoints:

- "/api/commissioning-service"

- name: "Reporting Service"

responsibilities:

- "Consolidate and export reports"

dependencies:

- "Workforce Management Service"
- "Project and Work Tracking Service"
- "CRM and Proposals Service"
- "Financial Management Service"
- "Billing Service"
- "Content and Document Management Service"
- "Commissioning Service"

suggested_endpoints:

- "/api/reporting-service"

- name: "Notification Service"

responsibilities:

- "Send email notifications"

dependencies:

- "Administration Service"

suggested_endpoints:

- "/api/notification-service"

- name: "File Storage Service"

responsibilities:

- "Store and retrieve files"

dependencies:

- "Administration Service"

suggested_endpoints:

- "/api/file-storage-service"

interactions:

- from: "Administration Service"

to: "contract types"

- from: "Identity and Access Service"

to: "contractual"

- from: "Workforce Management Service"

to: "Administration Service"

- from: "Workforce Management Service"

to: "Identity and Access Service"

- from: "Workforce Management Service"

to: "Notification Service"

- from: "Project and Work Tracking Service"

to: "Identity and Access Service"

- from: "Project and Work Tracking Service"

to: "Workforce Management Service"

- from: "Project and Work Tracking Service"

to: "File Storage Service"

- from: "Project and Work Tracking Service"

to: "Notification Service"

- from: "CRM and Proposals Service"

to: "contacts"

- from: "Financial Management Service"

to: "entries"

- from: "Billing Service"

to: "taxes"

- from: "Content and Document Management Service"

to: "Identity and Access Service"

- from: "Content and Document Management Service"

to: "File Storage Service"

- from: "Commissioning Service"

to: "Project and Work Tracking Service"

- from: "Commissioning Service"

to: "Workforce Management Service"

- from: "Commissioning Service"

to: "File Storage Service"

- from: "Commissioning Service"

to: "Notification Service"

- from: "Reporting Service"

to: "Workforce Management Service"

- from: "Reporting Service"

to: "Project and Work Tracking Service"

- from: "Reporting Service"

to: "CRM and Proposals Service"

- from: "Reporting Service"

to: "Financial Management Service"

- from: "Reporting Service"

to: "Billing Service"

- from: "Reporting Service"

to: "Content and Document Management Service"

- from: "Reporting Service"

to: "Commissioning Service"

- from: "Notification Service"

to: "Administration Service"

- from: "File Storage Service"

to: "Administration Service"